Flood situation report

Area: Teesta glacial lake flood, Sikkim, 4 Oct 2023 | Event date: 4 October 2023 | Latest radar image: 14 October 2023

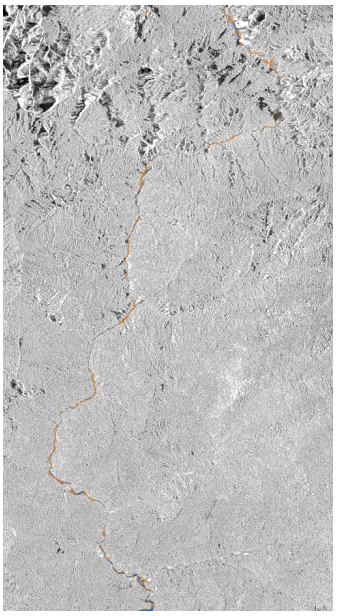
1. Where it flooded

Satellite images show 859 ha (8.59 km²) flooded or buried by debris. F010 F011

Of this, 147 ha is open water and 712 ha is debris or scour. F012 F014

The radar could not see 4,915 ha because of terrain shadow, so conditions there are unknown. F016

The places with the most buildings likely destroyed: Pegong, Sangkalang, Dikchu, Majitar, Simchuthang. F057



Radar change map, before and after the event.

2. What was hit

Buildings: 490 likely destroyed and 77 possibly damaged, out of 8,835 mapped. F025 F021 F022

Roads: 35.4 km of road lies under mapped debris or water; the road sections that touch it total 79.1 km, out of 2,268.0 km mapped. F040 F034 F038

Bridges: 18 likely destroyed (North Sikkim Highway, Sirwani Bridge) and 6 possibly damaged, out of 139 mapped. F031 F027 F028 F033

3. Who is cut off

9 of 41 settlements are likely cut off by road. F059 F058

Among them: Pegong, Dikchu, Lingthem, Passingdong, Bringbong. F063

The nearest health facility for 9 of the cut-off settlements is Mangan District Hospital. F064 F065

OpenStreetMap lists 99 health facilities in the area (hospitals, health posts and clinics); its labels are not verified. F323 F324

Repair priority

1. Reopening unnamed secondary bridge reconnects 7 settlements. F326 F327 F330
2. Reopening unnamed trunk (road) reconnects 5 settlements. F332 F333 F335
3. Reopening unnamed unclassified (road) reconnects 1 settlement. F337 F338 F340
4. Reopening North Sikkim Highway bridge reconnects 1 settlement. F342 F343 F346

Method

Sentinel-1 radar images from before and after the event are compared to map new water and debris. Pre-event OpenStreetMap roads, bridges and buildings are overlaid, and a road network model finds settlements that can no longer reach a health facility.

Educational prototype, not an operational tool. Numbers come from satellite analysis and have not been checked on the ground.

Grey codes such as F012 point to entries in facts.json.

Contains modified Copernicus Sentinel data 2026.
Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European Union and ESA; all rights reserved.
© OpenStreetMap contributors.